60 x 20 mm / 4 copper layers / black solder mask / 5 A at 24 V design target
4 MB flash, no PSRAM. USB powers 5 V strips only. Wired input has priority.
Initial fabrication design: assembled electrical, thermal and RF validation required.

ESP32-S3 / no PSRAM

File: mcu.kicad_sch

MCU power, straps and 40 MHz clock

File: core.kicad_sch

MCU power, straps and 40 MHz clock (continued)

File: core_2.kicad_sch

4 MB flash and SPI termination

File: flash.kicad_sch

2.4 GHz ceramic antenna and RF matching

File: rf.kicad_sch

USB-C data and source detection

File: usb.kicad_sch

Wired priority and USB current limiting

File: power.kicad_sch

Wired priority and USB current limiting (continued)

File: power_2.kicad_sch

3.3 V buck and 5 V logic boost

File: regulators.kicad_sch

3.3 V buck and 5 V logic boost (continued)

File: regulators_2.kicad_sch

Two 5 V data outputs and strip terminals

File: outputs.kicad_sch

Centered BOOT / GPIO18 button

File: button.kicad_sch

Production and recovery test pads

File: test.kicad_sch

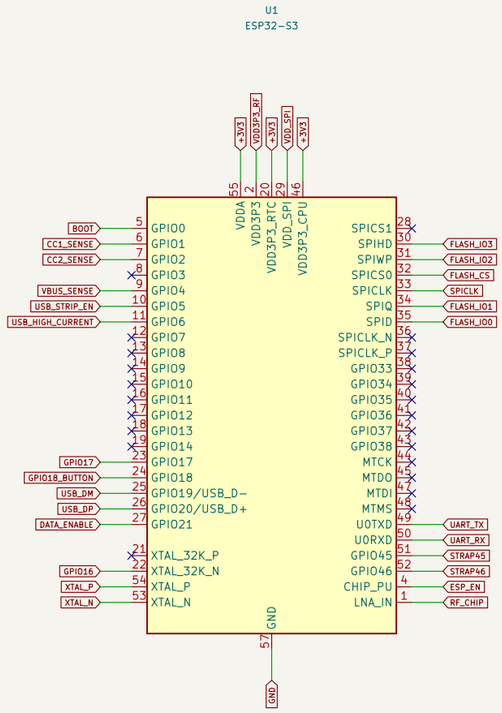
Luxio

Sheet: /
File: LX0001.kicad_sch

Title: LX0001 digital LED controller

Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
KiCad E.D.A. 10.0.7		Id: 1/14

Bare ESP32-S3, 4 MB external quad-SPI flash. GPIO16 / GPIO17 drive 5 V buffers.
GPIO18 senses the shared BOOT button through an isolation diode.



PWR_FLAG
GND

Luxio

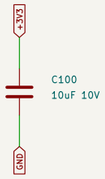
Sheet: /ESP32-S3 / no PSRAM/
File: mcu.kicad_sch

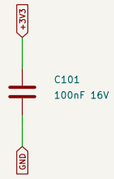
Title: ESP32-S3 / no PSRAM

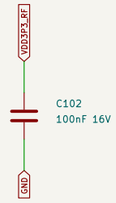
Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
KiCad E.D.A. 10.0.7		Id: 2/14

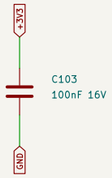
3.3 V power, local bypassing, boot straps and 40 MHz crystal.
GPIO45 and GPIO46 pulled LOW. Boot recovery: hold SW1 and pulse EN low.

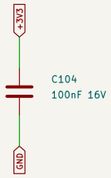


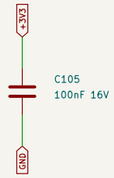


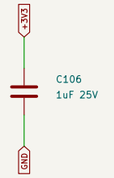


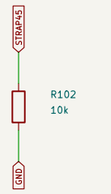


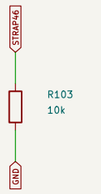


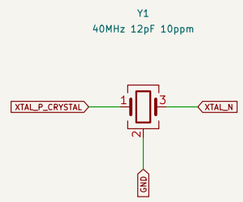


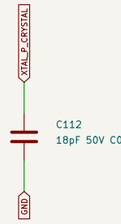
















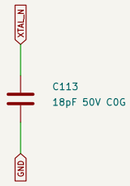
Luxio

Sheet: /MCU power, straps and 40 MHz clock/
File: core.kicad_sch

Title: MCU power, straps and 40 MHz clock

Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
KiCad E.D.A. 10.0.7		Id: 3/14

3.3 V power, local bypassing, boot straps and 40 MHz crystal.
GPIO45 and GPIO46 pulled LOW. Boot recovery: hold SW1 and pulse EN low.

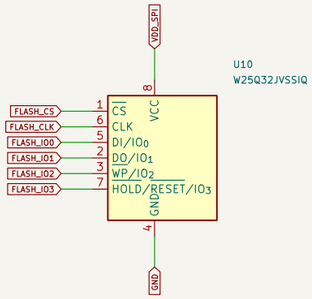


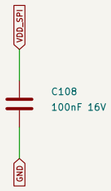


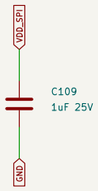
Luxio

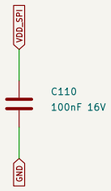
Sheet: /MCU power, straps and 40 MHz clock (continued)/		
File: core_2.kicad_sch		
Title: MCU power, straps and 40 MHz clock (continued)		
Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
KiCad E.D.A. 10.0.7		Id: 4/14

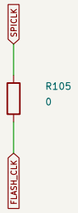
W25Q32JVSSIQ, 32 Mbit = 4 MB, no PSRAM.
VDD_SPI is the ESP32 flash supply. R105 reserves clock-series tuning.

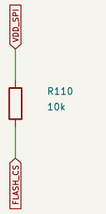














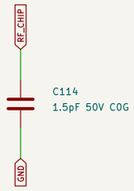
Luxio

Sheet: /4 MB flash and SPI termination/
File: flash.kicad_sch

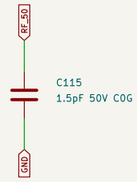
Title: 4 MB flash and SPI termination

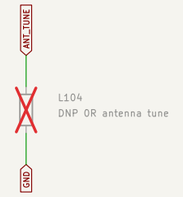
Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
KiCad E.D.A. 10.0.7		Id: 5/14

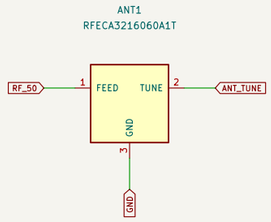
Walsin RFECA3216060A1T. Preserve the 8 x 3 mm edge clearance on ALL copper layers.
RF matching values are initial values; tune on assembled PCB in final enclosure.











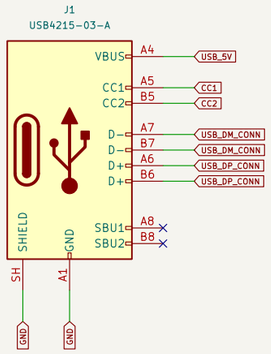
Luxio

Sheet: /2.4 GHz ceramic antenna and RF matching/
File: rf.kicad_sch

Title: 2.4 GHz ceramic antenna and RF matching

Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
KiCad E.D.A. 10.0.7		Id: 6/14

USB 2.0 device / 5 V sink; no PD. Separate 5.1k CC pull-downs.
Firmware must obey enumeration, suspend and Type-C advertised current.

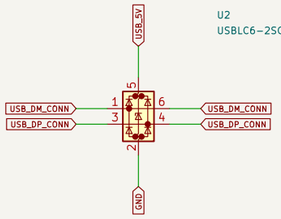


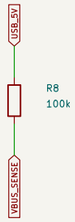


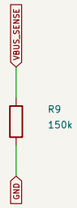




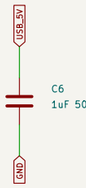


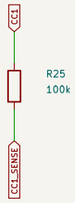


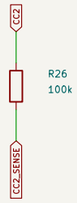




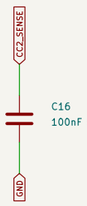














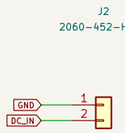
Luxio

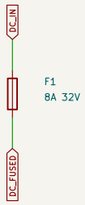
Sheet: /USB-C data and source detection/
File: usb.kicad_sch

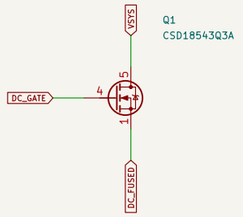
Title: USB-C data and source detection

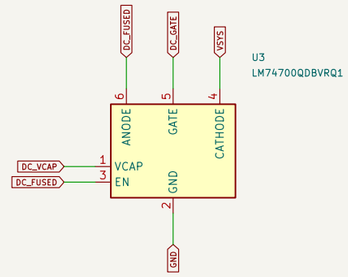
Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
KiCad E.D.A. 10.0.7		Id: 7/14

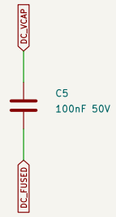
5–24 V regulated wired supply has hardware priority over USB.
Reverse-blocking paths isolate sources. Design target 5 A continuous wired output.

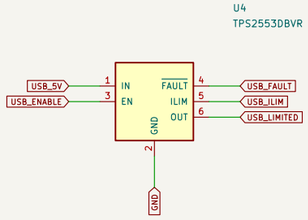


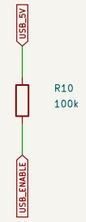


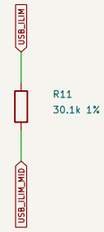








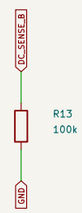


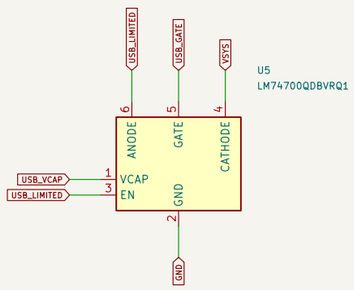


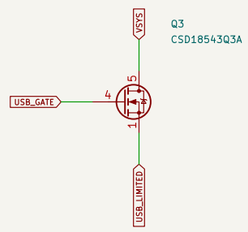


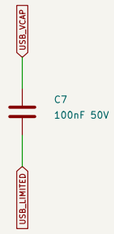


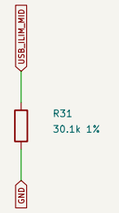


















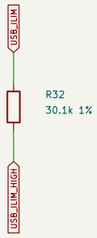
Luxio

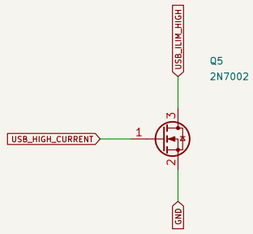
Sheet: /Wired priority and USB current limiting/
File: power.kicad_sch

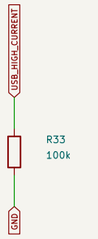
Title: Wired priority and USB current limiting

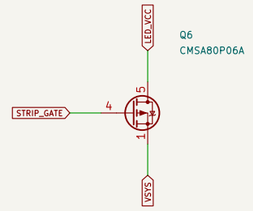
Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
KiCad E.D.A. 10.0.7		Id: 8/14

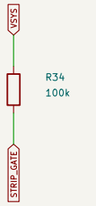
5–24 V regulated wired supply has hardware priority over USB.
Reverse-blocking paths isolate sources. Design target 5 A continuous wired output.



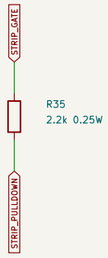


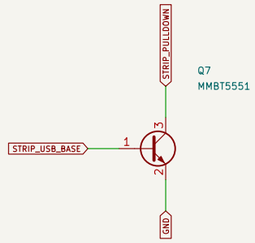


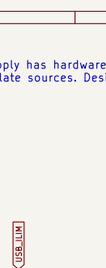


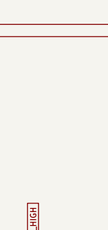


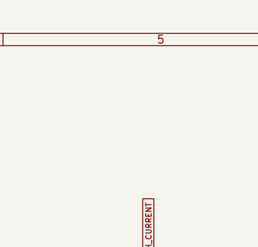


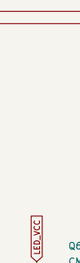


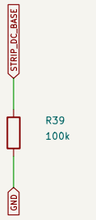


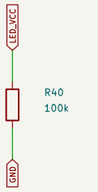












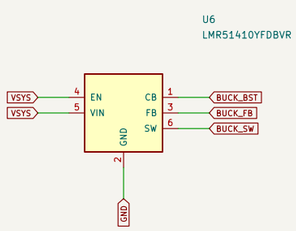
Luxio

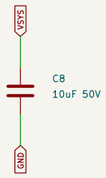
Sheet: /Wired priority and USB current limiting (continued)/
File: power_2.kicad_sch

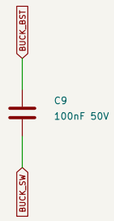
Title: Wired priority and USB current limiting (continued)

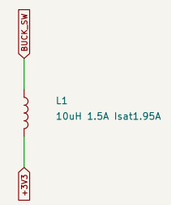
Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
KiCad E.D.A. 10.0.7		Id: 9/14

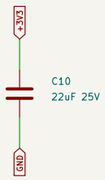
3.3 V buck from VSYS; boost 3.3 V to 4.99 V for AHCT buffers only.
Strip output follows selected input voltage; no 12/24 V boost from USB.

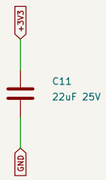


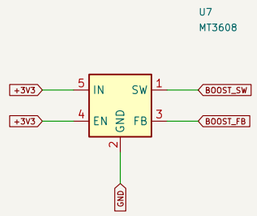


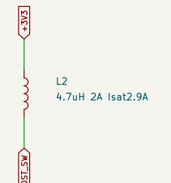


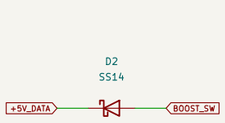


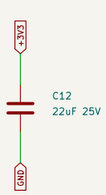


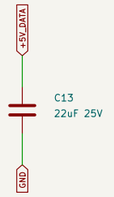


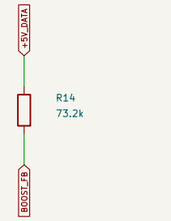


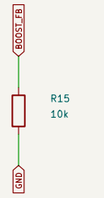


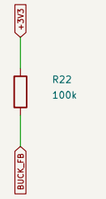


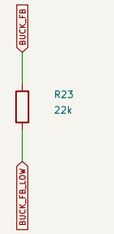


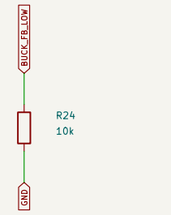














Luxio

Sheet: /3.3 V buck and 5 V logic boost/
File: regulators.kicad_sch

Title: 3.3 V buck and 5 V logic boost

Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
KiCad E.D.A. 10.0.7		Id: 10/14

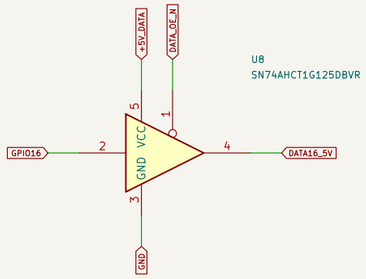
3.3 V buck from VSYS; boost 3.3 V to 4.99 V for AHCT buffers only.
Strip output follows selected input voltage; no 12/24 V boost from USB.



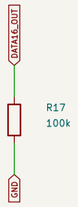
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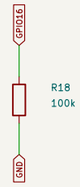
Sheet: /3.3 V buck and 5 V logic boost (continued)/		
File: regulators_2.kicad_sch		
Title: 3.3 V buck and 5 V logic boost (continued)		
Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
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GPIO16 / GPIO17 -> AHCT buffers -> 100-ohm series resistors.
DATA_ENABLE defaults low; set both data outputs low before enabling buffers.

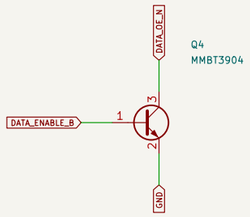


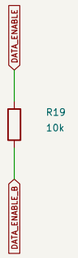






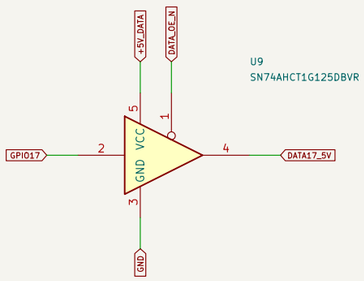


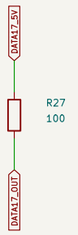


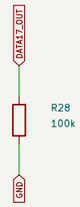


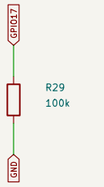




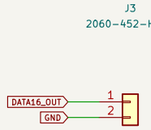


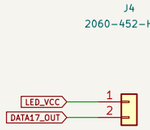












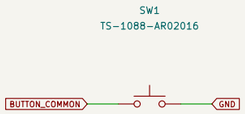
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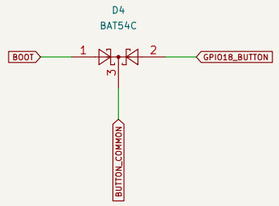
Sheet: /Two 5 V data outputs and strip terminals/
File: outputs.kicad_sch

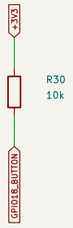
Title: Two 5 V data outputs and strip terminals

Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
KiCad E.D.A. 10.0.7		Id: 12/14

SW1 pulls BOOT and GPIO18 low through separate Schottky diodes.
Firmware uses GPIO18 as an input. BOOT must be held low during reset for download.







Luxio

Sheet: /Centered BOOT / GPIO18 button/ File: button.kicad_sch		
Title: Centered BOOT / GPIO18 button		
Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
KiCad E.D.A. 10.0.7		Id: 13/14

Test-point map for recovery, power rails and data.
Use common ground. Firmware/configuration needed for USB budget and LED enable.

TP1
GND

TP2
+3V3

TP3
ESP_EN

TP5
USB_FAULT

TP6
GPIO16

TP7
GPIO17

TP8
+5V_DATA

TP9
VSYS

TP10
GPIO18_BUTTON

TP11
UART_TX

TP12
UART_RX

Luxio

Sheet: /Production and recovery test pads/
File: test.kicad_sch

Title: Production and recovery test pads

Size: A3	Date: 2026-11-11	**Rev: 1.0.0**
KiCad E.D.A. 10.0.7		Id: 14/14